NOTE TO FILE: ROTOR FLANGE MACHINING WAS ADDED TO SCOPE OF WORK AFTER COMPLETION OF ROTOR WELD MODIFICATION (SEE ROTOR ASSEMBLY ITP I-GMS-6300 FOR THAT PORTION OF THE WORK).

ANDRITZ

Installation

INSPECTION AND TEST PLAN

Document #: I-GMS - 6301 Rotor flange machining

Project: GMS Poles Unit 1-4

Revision : 00

Unit #:

Codes: HC : hold point (customer)
WC : witness point (customer)
HA : hold point (Andritz Technical support)
WA : witness point (Andritz Technical support)

ITP step #	Code	Description	Procedure #	Procedure step #	Comments	Form	Date and initials
Preparation							
6301 - 1	WC	Rotor rim concentricity and rotor flange level	P-6300-03	1.5		Q-0000-001	22 & 25 July 2012
6301 - 2	WC	Milling machine alignment	P-6300-03	1.6		Q-0000-001	27/7/2012
Rotor flange machining							
6301 - 3	WC	Flange thickness	P-GMS-6300-99	2.1		Q-0000-001	27/9/2012
6301 - 4	WC	Rotor flange shape	P-GMS-6300-99	2.2	Before machining	Q-0000-001	21/7/12
6301 - 5	WC	Rotor flange shape	P-GMS-6300-99	2.6	After machining	Q-0000-001	25/7/12

ANDRITZ

Installation
INSPECTION AND TEST PLAN

Document #: I-GMS - 6301   Rotor flange machining   Revision : 00

Project: GMS Poles Unit 1-4   Unit #:

Codes: HC : hold point (customer)
WC : witness point (customer)
HA : hold point (Andritz Technical support)
WA : witness point (Andritz Technical support)

ITP step #	Code	Description	Procedure #	Procedure step #	Comments	Form	Date and initials
Thrust block							
6301 - 6	WC	Thrust block coupling and centering	P-GMS-6300-99	4.1		Q-0000-001	7/31/12
6301 - 7	WC	Contact surface assessment	P-GMS-6300-99	4.12 to 4.14		Q-0000-001	7/31/12
6301 - 8	WC	Welding visual inspection and PT inspection	P-6300-02A	4.16	STEP CANCELLED REF: FAM-BC4-AH-022	Q-0400-011 Q-0400-012.	2012-08-10

Revision 00
Prepared by: François Sergerie

Date: 18-Jul-2012

signed ITP

july 31 - GMS U4 ROTOR FINAL Q-0000-001 R_00.xlsx

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

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Revision # 00

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Unit : 4

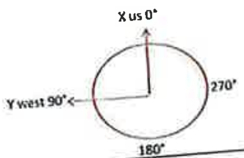
General Inspection Report

ITP step #: 6301-1

Project: GMS - rotor flange machining

instruction : Measure rim piling best center in relation to measured rim axis. 2 in from top and bottom of the 3 rim lamination piling at poles in front of supports

Procedure P-GMS-6300-097 Step 1.6, 2 2



C_CIR_BOTTOM_LOW_12_POLES (Constructed Circle , number of points 12)

X 0.003
Y 0.003

Diameter 413.393
Circularity 0.027

C_CIR_MIDDLE_HIGH_12_POLES (Constructed Circle , number of points 12)

0.003
0.006

Diameter 413.391
Circularity 0.014

C_CIR_BOTTOM_HIGH_12_POLES (Constructed Circle , number of points 12)

X 0.001
Y 0.003

Diameter 413.391
Circularity 0.016

C_CIR_TOP_LOW_12_POLES Constructed Circle , number of points 12)

0.004
0.011

Diameter 413.384
Circularity 0.017

C_CIR_MIDDLE_LOW_12_POLES (Constructed Circle , number of points 12)

X 0.001
Y 0.001

Diameter 413.386
Circularity 0.012

C_CIR_TOP_HIGH_12_POLES (Constructed Circle , number of points 12)

0.008
0.017

Diameter 413.418
Circularity 0.020

C_CYL_AXIS_POLES_(12) (Constructed Cylinder , number of points 71)

X 0
Y 0
Z Z -9.568
Diameter 413.394
Normal I 0.000
J 0.000
K 1.000
RMS 0.009
Standard Dev 0.009
Max 0.024
Min -0.018
Form 0.042

Step: After rim was moved back on thrust bloc. It is now sitting on short supports that are positioned on brake track

Time : All day

Date : july 20 2012

dd-mmm-yyy

NCR #: -

Measurement instruments
Laser tracker X0100502531

start
end

Measured by: P. Mire

ANDRITZ Rep : Mike Kelly

Cust. Repres : Kate Taylor

ANDRITZ Hydro

Installation and Commissioning
Quality Control Record

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Revision # 00
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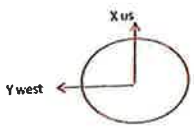
General Inspection Report

ITP step #: 6301-2
Project: GMS - rotor flange machining

instruction : During rotation of milling arm and record 16 points at it's extremity

Procedure P-GMS-6300-097 Step 1.6, 2.2



M_PL_MILLING_ROTATION_ARM_END *(Measured Plane , number of points 15)*

		Measured
Center	X	3.513
	Y	3.247
	Z	-13.912
Form		0.004

Readings

ID	Component	X (in)	Y (in)	**Z (in)**	Dev Fit (in)	Dev Pos	Laser Sight Orientation
1	Plane	38.631	-41.256	**-13.9127**	0.0023	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight
2	Plane	51.425	-23.463	**-13.9091**	-0.0012	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight
3	Plane	51.425	-23.463	**-13.9090**	-0.0014	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight
4	Plane	56.459	-2.777	**-13.9105**	-0.0001	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight
5	Plane	53.938	16.925	**-13.9108**	-0.0002	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight
6	Plane	42.683	37.077	**-13.9117**	0.0002	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight
7	Plane	27.478	49.421	**-13.9137**	0.0018	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight
8	Plane	4.053	56.418	**-13.9121**	-0.0004	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight
9	Plane	-13.602	54.916	**-13.9121**	-0.0008	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight
10	Plane	-35.793	42.985	**-13.9128**	-0.0003	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight
11	Plane	-55.710	9.859	**-13.9129**	0.0001	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight
12	Plane	-56.208	-6.410	**-13.9136**	0.0008	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight
13	Plane	-47.663	-30.464	**-13.9137**	0.0000	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight
14	Plane	-40.697	-39.283	**-13.9117**	-0.0003	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight
15	Plane	-22.727	-51.783	**-13.9111**	-0.0003	C_DEVICEPOS5_-_MILLING_SETUP	Front Sight

max	-13.909
min	-13.914
flatness	0.005

NOTE : Vibrations where prensent in the plant. Indicator showed vibration reaching .001 in (approx). Check was also performed on milling quil, the difference in elevation on possible points either side showed .0007in difference

Machining step: At milling set up
Time : Afternoon
Date : july 22 2012

NCR #: -

dd-mmm-yyyy

Measurement Instruments		T°	
Laser tracker X0100802531	Before:	23.3 °C	
	After:	24 °C	

Machining step: Prior to machining

ANDRITZ Rep.: Mike Kelly
Cust. Repres.: Katie Taylor

ANDRITZ Hydro

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ITP step #: 6301-3

Project: GM Shrum

Unit : 4

Equipment : Thickness of rotor coupling flange before machining

#	ID	OD
1	3.466	3.481
2	3.463	3.478
3	3.463	3.483
4	3.461	3.479
5	3.466	3.478
6	3.467	3.479
7	3.467	3.479
8	3.469	3.477
9	3.464	3.487
10	3.465	3.478
11	3.465	3.477
12	3.470	3.475
13	3.465	3.475
14	3.467	3.481
15	3.468	3.489
16	3.470	3.479

Avg	3.466	3.480
Max	3.470	3.489
Min	3.461	3.475

Note: * Measurement are taken in coupling holes using a straight edge and dept mic.

NCR #:

Measurement Instruments	T°	Measured by:	dd-mmm-yyyy
dept micrometer	Before: 23.3	D. Zaharoff	22-Jul-2012
#417050	After: 23.3	D.MacCormack	
		ANDRITZ Rep.: Mike Kelly	
		Cust. Repres.	

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ANDRITZ Hydro

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General Inspection Report

ITP step #: 6301-4
Project: GMS - rotor flange machining

Unit : 4

instruction : Mesure 16 points 2 in from edge of flange ID and OD

Procedure P-GMS-6300-097 Step 1.6, 2.2

X us

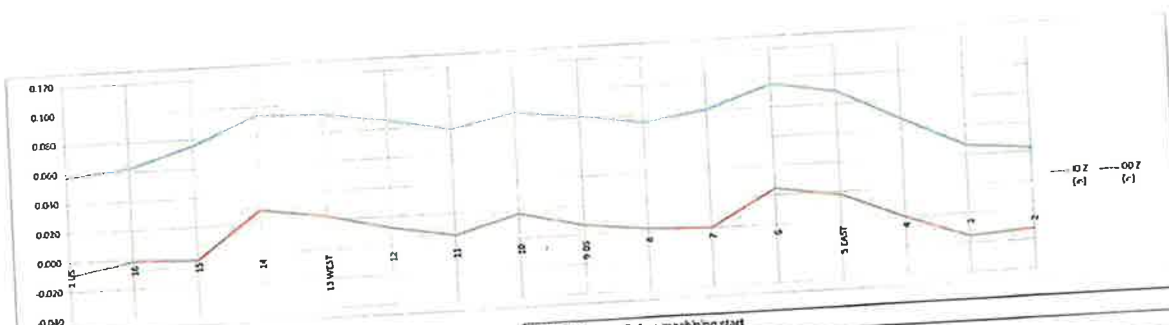
Y west

	ID				
	X (in)	Y (in)	radius (in)	angle (deg)	ID Z (in)
1 US	[illegible]	[illegible]	[illegible]	5	0.058
16	[illegible]	[illegible]	[illegible]	[illegible]	0.063
15	[illegible]	[illegible]	[illegible]	[illegible]	0.076
14	[illegible]	[illegible]	[illegible]	[illegible]	0.094
13 WEST	[illegible]	[illegible]	[illegible]	[illegible]	0.092
12	[illegible]	[illegible]	[illegible]	[illegible]	0.086
11	[illegible]	[illegible]	[illegible]	[illegible]	0.078
10	[illegible]	[illegible]	[illegible]	[illegible]	0.088
9 DS	[illegible]	[illegible]	[illegible]	[illegible]	0.081
8	[illegible]	[illegible]	[illegible]	[illegible]	0.074
7	[illegible]	[illegible]	[illegible]	[illegible]	0.081
6	[illegible]	[illegible]	[illegible]	[illegible]	0.096
5 EAST	[illegible]	[illegible]	[illegible]	[illegible]	0.069
4	[illegible]	[illegible]	[illegible]	[illegible]	0.067
3	[illegible]	[illegible]	[illegible]	[illegible]	0.047
2	[illegible]	[illegible]	[illegible]	[illegible]	0.043

max 0.096
min 0.043
flatness 0.053

	OD					od to id angle
	X (in)	Y (in)	radius (in)	angle (deg)	OD Z (in)	
1 US	[illegible]	[illegible]	[illegible]	[illegible]	-0.009	0.4
16	[illegible]	[illegible]	[illegible]	[illegible]	-0.001	0.4
15	[illegible]	[illegible]	57.0	[illegible]	-0.003	0.5
14	[illegible]	[illegible]	[illegible]	[illegible]	0.020	0.4
13 WEST	[illegible]	[illegible]	[illegible]	[illegible]	0.023	0.4
12	23.522	[illegible]	[illegible]	[illegible]	0.012	0.4
11	[illegible]	[illegible]	[illegible]	[illegible]	0.005	0.4
10	[illegible]	[illegible]	[illegible]	[illegible]	0.017	0.4
9 DS	[illegible]	[illegible]	[illegible]	[illegible]	0.007	0.5
8	[illegible]	[illegible]	[illegible]	[illegible]	0.002	0.4
7	[illegible]	[illegible]	[illegible]	[illegible]	0.000	0.5
6	[illegible]	[illegible]	[illegible]	[illegible]	0.024	0.4
5 EAST	[illegible]	[illegible]	[illegible]	[illegible]	0.018	0.4
4	[illegible]	[illegible]	[illegible]	[illegible]	0.000	0.4
3	[illegible]	[illegible]	[illegible]	[illegible]	-0.015	0.3
2	[illegible]	[illegible]	[illegible]	[illegible]	-0.012	0.3

max 0.029
min -0.015
flatness 0.045



Machining step: Before machining start
Time : Afternoon
Date : july 21 2012

NCR #: _

Measurement Instruments		T°
Laser tracker X0100802531	Before:	24.6 °C
	After:	24 °C

		dd-mm-yyyy
Measured by:	P. Mitre, D. McCromac, D. Zaharoff	
ANDRITZ Rep :	Mike Kelly	
Cust. Repres :	Katie Taylor, Dany	

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

General Inspection Report

Q-0000-001

ITP step #: 6301-4

Project: GMS - rotor flange machining

Unit : 4

Revision # 00

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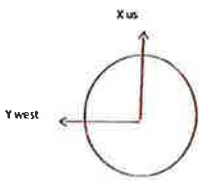
instruction : Mesure 16 points 1 in from edge of flange ID and OD on machined surface

Procedure P-GMS-6300-097 Step 1.6, 2.2



ID	X (in)	Y (in)	radius (in)	angle (deg)	ID Z (in)
1 US	54.071	5.073	54.3	5	1.013
16	49.823	24.022	55.3	26	1.011
15	37.946	40.528	55.5	47	1.010
14	20.102	53.816	57.4	70	1.007
13 WEST	-2.561	57.135	57.2	93	1.008
12	-19.853	52.980	56.6	111	1.010
* 11					
10					
9 DS	-55.684	-2.914	55.7	183	1.011
8	-49.673	-24.170	55.2	206	1.011
7	-38.055	-41.216	56.1	227	1.009
6	-20.112	-53.341	57.0	249	1.010
5 EAST	4.298	-56.345	56.5	274	1.011
4	21.260	-50.431	54.7	293	1.012
3	40.905	-35.188	54.0	319	1.012
2	51.679	-16.693	54.3	342	1.014

max 1.014

min 1.007

flatness 0.007

OD	X (in)	Y (in)	radius (in)	angle (deg)	OD Z (in)
1 US	57.491	2.550	57.5	3	1.014
16	52.316	24.878	57.9	25	1.010
15	39.805	42.645	58.3	47	1.009
14	20.363	54.928	58.6	70	1.007
13 WEST	-2.768	58.316	58.4	93	1.007
12	-20.061	54.376	58.0	110	1.009
* 11					
10	-55.825	18.573	58.8	162	1.012
9 DS	-58.087	-3.343	58.2	183	1.011
8	-52.106	-25.352	57.9	206	1.010
7	-39.658	-43.592	58.9	228	1.007
6	-20.565	-54.917	58.6	249	1.010
5 EAST	4.588	-58.344	58.5	274	1.010
4	23.051	-53.453	58.2	293	1.010
3	43.481	-38.351	58.0	319	1.012
2	54.713	-19.003	57.9	341	1.013

max 1.014

min 1.007

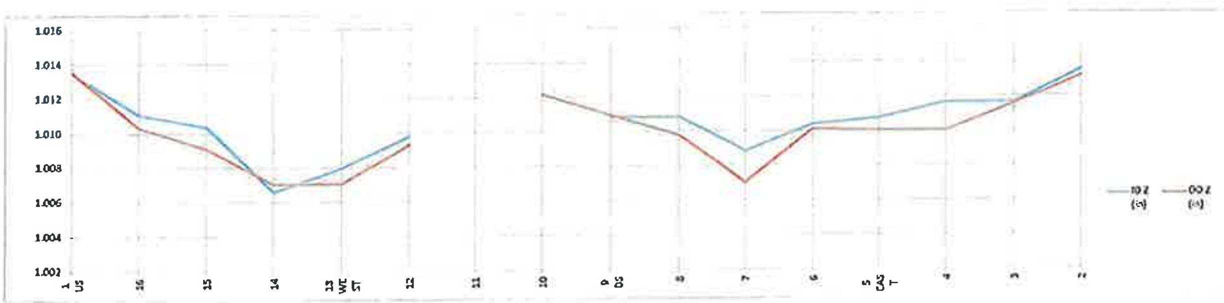
flatness 0.007

average perp (in)	pt
0.003	1-9
0.002	16-8
0.000	15-7
0.003	14-6
0.003	13-5
0.002	14-4
	13-3
0.001	12-2

average

0.003	perpendicularity

NOTE: * coldnt be recorded, milling was in the way



Machining step: After pass 1 machining. Took off .042 in @ on flange OD (PT3)
Time : Afternoon
Date : july 23 2012

NCR #: -

Measurement Instruments		T°	Measured by:	P. Mitro, D. Maccormac	dd-mmm-yyyy
Laser tracker X0100802531	Before:	24.5 °C			
	After:	27.5 °C	ANDRITZ Rep :	Mike Kelly	
			Cust. Repres :	Katie Taylor	

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Q-0000-001

Revision # 00

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General Inspection Report

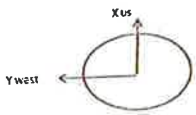
ITP step #: 6301-4

Project: GMS - rotor flange machining

Unit : 4

Instruction : Mesure 16 points 1 in from edge of flange ID and OD on machined surface

Procedure P-GMS-6300-097 Step 1.6, 2.2



	ID				
	X (in)	Y (in)	radius (in)	angle (deg)	ID Z (in)
1 US	53.504	5.040	53.7	5	0.820
16	47.828	25.192	54.1	28	0.819
15	38.915	38.338	54.6	45	0.818
14	15.356	52.637	54.8	74	0.815
13 WEST	-5.717	54.349	54.6	96	0.817
12	-20.394	51.159	55.1	112	0.818
* 11					
10	-51.986	16.196	54.5	163	0.821
9 DS	-54.064	-0.648	54.1	181	0.819
8	-47.498	-25.972	54.1	209	0.818
7	-38.957	-38.040	54.4	224	0.818
6	-15.088	-52.029	54.2	254	0.819
5 EAST	4.580	-54.249	54.4	275	0.819
4	25.727	-47.743	54.2	298	0.818
3	38.458	-37.971	54.0	315	0.819
2	51.741	-16.500	54.3	342	0.820

max 0.821

min 0.815

flatness 0.005

	OD				
	X (in)	Y (in)	radius (in)	angle (deg)	OD Z (in)
1 US	58.003	5.638	58.3	6	0.820
16	51.272	26.853	57.9	28	0.819
15	41.492	40.458	58.0	44	0.818
14	16.256	55.739	58.1	74	0.814
13 WEST	-5.534	57.906	58.2	95	0.817
12	-21.126	54.151	58.1	111	0.818
* 11					
10	-55.495	17.890	58.3	162	0.819
9 DS	-57.818	-1.293	57.8	181	0.819
8	-50.835	-28.488	58.3	209	0.819
7	-41.304	-40.912	58.1	225	0.819
6	-15.984	-55.454	57.7	254	0.819
5 EAST	5.177	-57.579	57.8	275	0.819
4	26.274	-51.228	57.6	297	0.819
3	41.720	-40.144	57.9	316	0.819
2	55.571	-17.606	58.3	342	0.821

max 0.821

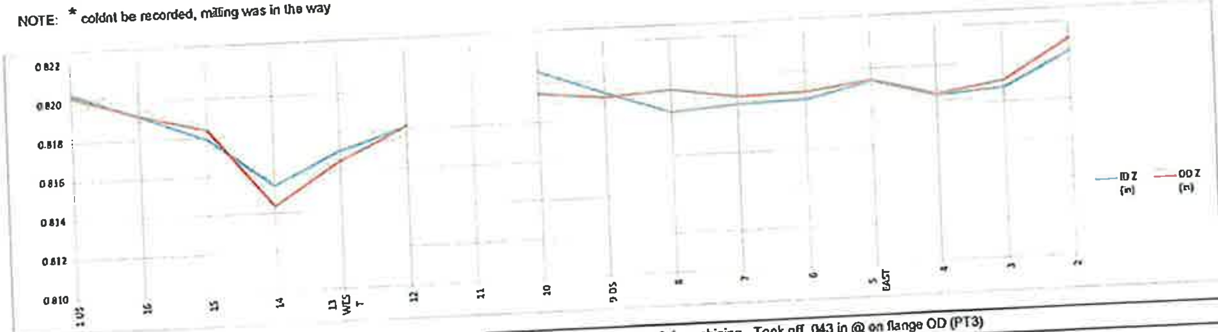
min 0.814

flatness 0.007

average perp (in)	pl
0.001	1-9
0.001	16-8
0.001	15-7
0.004	14-6
0.002	13-5
0.001	14-4
	13-3
0.000	12-2

average perpendicularity: 0.004

NOTE: * coldnt be recorded, milling was in the way



Machining step: After pass 2.1 machining. Took off .043 in @ on flange OD (PT3)	
Time : morning	
Date : july 24 2012	

NCR #: -

Measurement Instruments		T°	Measured by:	P. Mitre, D. Maccormac	dd-mm-yyyy
Laser tracker X0100802531	Before	24.5 °C	ANDRITZ Rep :	Mike Kelly	
	After.	25 °C	Cust. Repres.:	Katie Taylor	

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

General Inspection Report

Q-0000-001

ITP step #: 6301-4

Revision # 00

Project: GMS - rotor flange machining

Unit : 4

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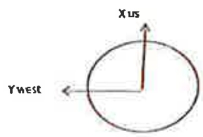
instruction : Mesure 16 points 1 in from edge of flange ID and OD on machined surface

Procedure P-GMS-6300-097 Step 1.6, 2.2



	ID				
	X (in)	Y (in)	radius (in)	angle (deg)	ID Z (in)
1 US	48.120	4.694	48.3	6	0.820
16	44.617	23.595	50.5	28	0.817
15	35.474	34.656	49.6	44	0.817
14	15.352	51.039	53.3	73	0.814
13 WEST	-4.609	51.866	52.1	95	0.815
12	-12.458	50.631	52.1	104	0.817
* 11					
10	-49.329	14.001	51.3	164	0.820
9 DS	-50.135	-0.650	50.1	181	0.819
8	-44.226	-24.392	50.5	209	0.819
7	-34.567	-38.648	51.9	228	0.817
6	-15.473	-50.931	53.2	253	0.818
5 EAST	5.264	-51.471	51.7	276	0.818
4	23.156	-43.218	49.0	298	0.819
3	34.520	-36.481	50.2	313	0.819
2	46.626	-13.982	48.7	343	0.821

max 0.821

min 0.814

flatness 0.007

	OD				
	X (in)	Y (in)	radius (in)	angle (deg)	OD Z (in)
1 US	55.077	4.816	55.3	5	0.819
16	49.686	26 292	56.2	28	0.817
15	39.392	39.248	55.6	45	0.817
14	16.646	54.110	56.6	73	0.813
13 WEST	-4.508	56.978	57.2	95	0.816
12	-21.978	52.744	57.1	113	0.819
• 11					
10	-55.015	15.331	57.1	164	0.819
9 DS	-56.681	-0.262	56.7	180	0.819
8	-51.302	-25.470	57.3	206	0.817
7	-37.589	-42.044	56.4	228	0.818
6	-19.044	-53.880	57.1	251	0.818
5 EAST	6.411	-57.187	57.5	276	0.819
4	26.079	-50.734	57.0	297	0.818
3	40.784	-39.559	56.8	316	0.818
2	52.675	-18.841	55.9	340	0.820

max 0.820

min 0.813

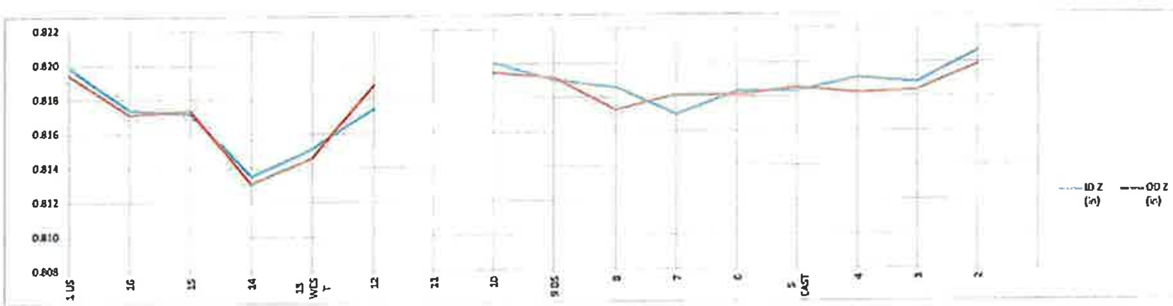
flatness 0.007

average perp (in)	pt
0.001	1-9
0.001	16-8
0.000	15-7
0.005	14-6
0.003	13-5
0.001	14-4
	13-3
0.001	12-2

average

0.005 perpendicularity

NOTE: * coldnt be recorded, milling was in the way



Machining step: After pass 3 machining. Cleaned up id to pass 2.1 level
Time : afternoon
Date : july 24 2012

NCR #: -

Measurement Instruments		T°	Measured by:	P. Mitre, T. Tomjenovic	dd-mmm-yyyy
Laser tracker X0100802531	Before:	25 °C			
	After:	24.7 °C	ANDRITZ Rep :	Mike Kelly, J-L Côté	
			Cust. Repres.:	Katie Taylor, Danny Burggraeve	

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ITP step #: 6301-4

Project: GMS - rotor flange machining

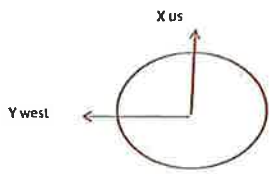
Unit : 4

Revision # 00

Page 1 of 1

instruction : record untouche zone after 3rd pass machining

Procedure P-GMS-6300-097 Step 1.6, 2.2



	X (in)	Y (in)	radius (in)	angle (deg)	Z (in)	left to reach 1/2in from ID (in)
1 US						0.0
16						0.0
15	32.056	33.632	46.5	46	0.829	2.0
14	15.720	43.670	46.4	70	0.847	5.0
13 WEST	-1.785	46.418	46.5	92	0.846	4.5
12	-11.307	44.997	46.4	104	0.848	3.5
11	*	*	*	*	*	2.5
10	-45.491	9.742	46.5	168	0.834	3.0
9 DS	-46.615	-1.148	46.6	181	0.833	2.0
8	-42.095	-19.906	46.6	205	0.831	2.5
7	-31.069	-34.625	46.5	228	0.836	3.0
6	-15.213	-44.010	46.6	251	0.849	4.5
5 EAST	2.407	-46.493	46.6	273	0.841	3.5
4	20.691	-42.091	46.9	296	0.821	1.5
3						0.0
2						0.0

	max	0.849	5.000
Machining step:	min	0.821	0.000
	flatness	0.029	
			2.3 average

average Z coordinate on pass 3 is: 0.818 in

NOTE: * coldnt be recorded, milling was in the way.

Machining step: After pass 3 ID machining to bring ID to OD level.
Time : Afternoon
Date : july 24 2012

NCR #: -

Measurement Instruments		T°			dd-mmm-yyyy
Laser tracker X0100802531	Before:	25 °C	Measured by:	P. Mitre, T. Tomjenovic	
Measuring tape	After:	24.7 °C	ANDRITZ Rep.:	Mike Kelly, J-L Côté	
			Cust. Repres.:	Katie Taylor, Danny Burggraeve	

ANDRITZ Hydro

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Q-0000-001

ITP step #: 6301-5

Revision # 00

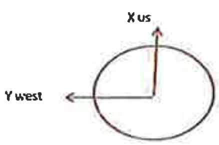
Project: GMS - rotor flange machining

Unit : 4

Page 1 of 1

instruction : Mesure elevation in 16 points 2 in from edge of flange ID and OD

Procedure P-GMS-6300-097 Step 1.6, 2.2



	ID				
	X (in)	Y (in)	radius (in)	angle (deg)	ID Z (in)
1 US	48.202	0.548	48.2	1	0.849
16	42.700	21.900	48.0	27	0.849
15	32.754	34.787	47.8	47	0.848
14	13.018	46.584	48.3	74	0.846
13 WEST	-5.748	47.264	47.6	97	0.847
12	-13.817	46.216	48.2	107	0.848
11					
10	-47.548	8.629	48.3	170	0.849
9 DS	-47.884	-3.740	48.0	184	0.848
8	-42.807	-22.118	48.2	207	0.848
7	-34.232	-33.487	47.9	224	0.847
6	-14.405	-45.976	48.2	253	0.847
5 EAST	3.715	-47.813	48.0	274	0.848
4	21.765	-42.771	48.0	297	0.848
3	34.559	-33.547	48.2	316	0.848
2	46.170	-14.497	48.4	343	0.850

max 0.850

min 0.846

flatness 0.004

	OD					average perp (in)	pt
	X (in)	Y (in)	radius (in)	angle (deg)	OD Z (in)		
1 US	57.119	-0.751	57.1	1	0.848	0.001	1-9
16	49.553	28.272	57.1	30	0.846	0.000	16-8
15	39.077	41.539	57.0	47	0.846	0.001	15-7
14	17.156	54.751	57.4	73	0.842	0.003	14-6
13 WEST	-4.940	56.754	57.0	95	0.844	0.002	13-5
12	-20.728	53.163	57.1	111	0.845	0.001	14-4
11							13-3
10	-54.375	16.724	56.9	163	0.848	0.001	12-2
9 DS	-55.858	-4.612	56.0	185	0.846		
8	-50.122	-26.236	56.6	208	0.847		
7	-40.868	-39.172	56.6	224	0.847		
6	-16.885	-54.308	56.9	253	0.844		
5 EAST	5.404	-56.730	57.0	275	0.846		
4	26.348	-50.787	57.2	297	0.847		
3	40.160	-40.062	56.7	315	0.847		
2	54.571	-16.531	57.0	343	0.848		

max 0.848

min 0.842

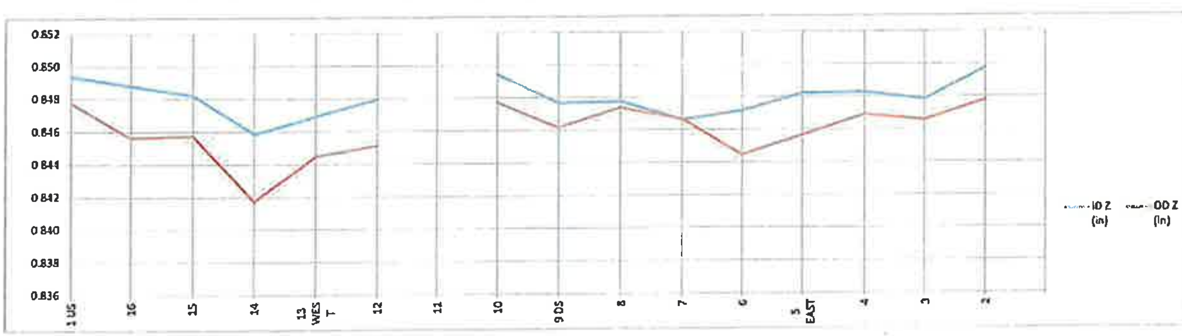
flatness 0.006

average

0.003 perpendicularity

Readings where taken with milling on rotor for all points except pt 11

BCH senior engineer Dany Burggraeve accepted untouched zones on the maching area. FAM #



Machining step: Pass 4 with milling still on rotor. Removed total .113 in @ pt3	
Time : 7 to 14h	
Date : july 25 2012	

NCR #: -

Measurement Instruments		T°	Measured by:	P. Mitre, D. Maccormac	dd-mmm-yyyy
Laser tracker X0100802531	Before:	25.6 °C			
	After:	29.7 °C	ANDRITZ Rep.:	Mike Kelly, J-L Côté	
			Cust. Repres :	Katie Taylor, Danny Burggraeve	

ANDRITZ Hydro

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Q-0000-001

ITP step #: 6301-5

Revision # 00

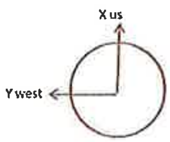
Project: GMS - rotor flange machining

Unit : 4

Page 1 of 1

instruction : Mesure elevation in 16 points 2 in from edge of flange ID and OD

Procedure P-GMS-6300-097 Step 1.6, 2.2



	ID				
	X (in)	Y (in)	radius (in)	angle (deg)	ID Z (in)
1 US	47.847	0.838	47.9	1	0.851
16	41.743	23.190	47.8	29	0.848
15	32.650	34.651	47.6	47	0.848
14	12.900	46.405	48.2	74	0.845
13 WEST	-3.990	47.894	48.1	95	0.847
12	-21.503	42.723	47.8	117	0.848
11	-33.275	34.905	48.2	134	0.847
10	-45.837	14.924	48.2	162	0.849
9 DS	-47.460	-5.434	47.8	187	0.848
8	-42.749	-22.253	48.2	207	0.847
7	-33.204	-34.726	48.0	226	0.847
6	-14.799	-45.605	47.9	252	0.847
5 EAST	5.936	-47.400	47.8	277	0.848
4	23.390	-41.983	48.1	299	0.849
3	33.708	-34.502	48.2	314	0.849
2	46.035	-14.476	48.3	343	0.851

max 0.851

min 0.845

flatness 0.006

	OD				
	X (in)	Y (in)	radius (in)	angle (deg)	OD Z (in)
1 US	57.118	1.373	57.1	1	0.849
16	49.686	27.866	57.0	29	0.846
15	38.488	41.996	57.0	47	0.846
14	15.080	54.977	57.0	75	0.842
13 WEST	-5.060	56.857	57.1	95	0.845
12	-25.908	50.926	57.1	117	0.846
11	-40.128	41.625	57.8	134	0.845
10	-54.455	16.537	56.9	163	0.848
9 DS	-56.569	-7.563	57.1	188	0.845
8	-50.000	-27.449	57.0	209	0.845
7	-41.239	-39.303	57.0	224	0.846
6	-16.710	-54.499	57.0	253	0.845
5 EAST	4.663	-57.137	57.3	275	0.846
4	26.646	-50.584	57.2	298	0.847
3	40.286	-40.538	57.2	315	0.847
2	54.353	-16.924	56.9	343	0.849

max 0.849

min 0.842

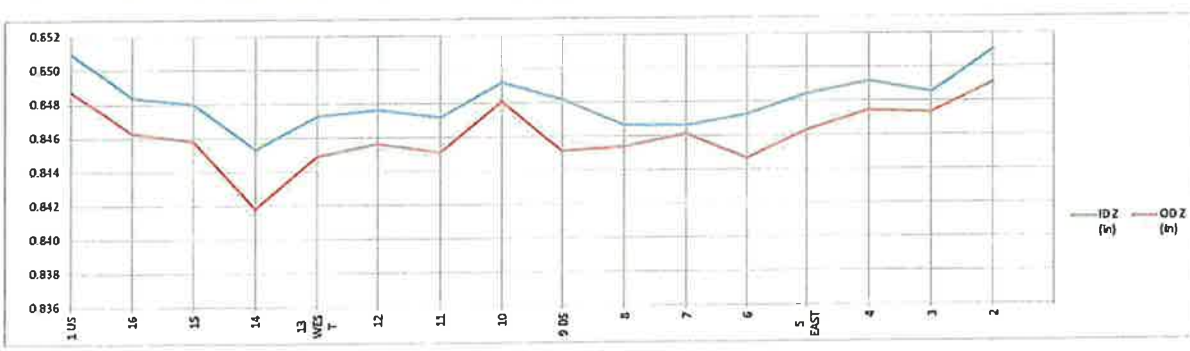
flatness 0.007

average perp (in)	pt
0.003	1-9
0.001	16-8
0.001	15-7
0.003	14-6
0.002	13-5
0.002	14-4
0.003	13-3
0.002	12-2

average perpendicularity: 0.003

Milling removed

BCH senior engineer Dany Burggraeve accepted untouched zones on the maching area. FAM #



Machining step: Pass 4 with milling removed, total mat'l removed @ PT3 : .113 in
Time : Afternoon
Date : July 25 2012

NCR #: -

Measurement Instruments		T°	Measured by:	P. Mitre, D. Maccormac	dd-mmm-yyyy
Laser tracker X0100802531	Before:	25.8 °C			
	After:	30 °C	ANDRITZ Rep :	Mike Kelly, J-L Côté	
			Cust. Repres :	Katie Taylor, Danny Burggraeve	

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ITP step #: 6301-6

Revision # 00

Project: GMS - rotor flange machining

Unit : 4

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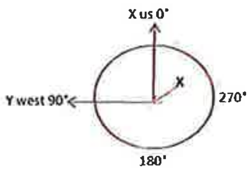
instruction : Measure thrust bloc centering to rim axis in 16 point with 2 circles. Measurements take place 6 1/2in below th bloc coupling flange

Procedure P-GMS-6300-097 Step 1.6, 2.2



M_CI_CENTER_MOVE1_ *(Measured Circle , number of points 16)*

Measured

Center	X	**0.0023**
	Y	**-0.0036**
	Z	0.0000
Diameter		118.256
Normal	I	0.00000000
	J	0.00000000
	K	1.00000000
RMS		0.0009
Standard Dev		0.0009
Max		0.0014
Min		-0.0011
Form		0.0026

Th block to rim axis concentricity:	0.0043
angle	302

	X	Y	rad (in)	angle (deg)	Z	Dev Fit
1 US	59.099	1.881	59.129	2	0.000	-0.001
16	53.358	25.476	59.128	26	0.000	0.000
15	40.254	43.309	59.127	47	0.000	0.001
14	19.972	55.651	59.127	70	0.000	0.001
13 WEST	-2.733	59.062	59.125	93	0.000	0.001
12	-25.321	53.427	59.123	115	0.000	0.000
11	-44.981	38.369	59.123	140	0.000	-0.001
10	-55.502	20.375	59.123	160	0.000	-0.001
9 DS	-59.092	-1.980	59.125	182	0.000	-0.001
8	-53.688	-24.770	59.127	205	0.000	0.000
7	-39.329	-44.154	59.130	228	0.000	0.001
6	-20.378	-55.509	59.132	250	0.000	0.001
5 EAST	4.075	-58.992	59.133	274	0.000	0.001
4	25.288	-53.452	59.132	295	0.000	0.000
3	43.890	-39.625	59.131	318	0.000	-0.001
2	55.482	-20.447	59.130	340	0.000	-0.001

Step: After move 1 on thrust bloc, light tight with impact gun
Time : End of day
Date : July 29 2012

NCR #: -

Measurement Instruments		T°	Measured by:	P. Mitre	dd-mmm-yyyy
Laser tracker X0100802531	Before:	24.7 °C			
	After:	27 °C	ANDRITZ Rep.:	Mike Kelly	
			Cust. Repres.:	Katie Taylor	

General Inspection Report

ITP step #: 6301-6

Project: GMS - rotor flange machining

Q-0000-001

Revision # 00

Unit : 4

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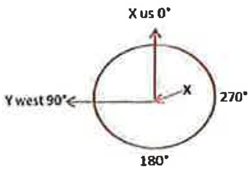
Instruction : Measure thrust bloc centering to rim axis in 16 point with 2 circles. Measurements take place 6 1/2in below th bloc coupling flange

Procedure P-GMS-6300-097 Step 1.6, 2.2



M_CI_TH_BL_FINAL_TH_BLOC *(Measured Circle , number of points 16)*

Measured

Center	X	0.001
	Y	-0.003
	Z	0.000
Diameter		118.256
	I	0
Normal	J	0
	K	1
RMS		0.0004
Standard Dev		0.0005
Max		0.0009
Min		-0.0009
Form		0.0018

Th block to rim axis concentricity:	0.0034
angle	290

	X	Y	rad (in)	angle (deg)	Z	Dev Fit
1 US	59.094	2.030	59.129	2	0	-0.001
16	53.462	25.258	59.128	25	0	0.000
15	40.861	42.735	59.127	46	0	0.000
14	20.304	55.530	59.126	70	0	0.000
13 WEST	-2.684	59.064	59.125	93	0	0.000
12	-26.018	53.571	59.125	115	0	0.000
11	-43.506	40.037	59.125	137	0	0.000
10	-55.465	20.482	59.126	160	0	0.000
9 DS	-59.104	-1.658	59.128	182	0	0.001
8	-54.060	-23.953	59.129	204	0	0.001
7	-39.143	-44.317	59.129	229	0	-0.001
6	-19.721	-55.745	59.130	251	0	-0.001
5 EAST	2.614	-59.074	59.132	273	0	0.000
4	26.668	-52.777	59.132	297	0	0.000
3	43.756	-39.774	59.131	318	0	0.000
2	55.488	-20.431	59.130	340	0	0.000

Step: Torquing completed
Time : Mid day to before break following day
Date : July 30 & 31 2012

NCR #: -

Measurement Instruments		T°	Measured by:	P. Mitre	dd-mmm-yyyy
Laser tracker X0100802531	day 1	24.8 to 25 °C			
	day 2	25.1 to 25.3 °C	ANDRITZ Rep :	Mike Kelly	
			Cust. Repres.:	Katie Taylor	

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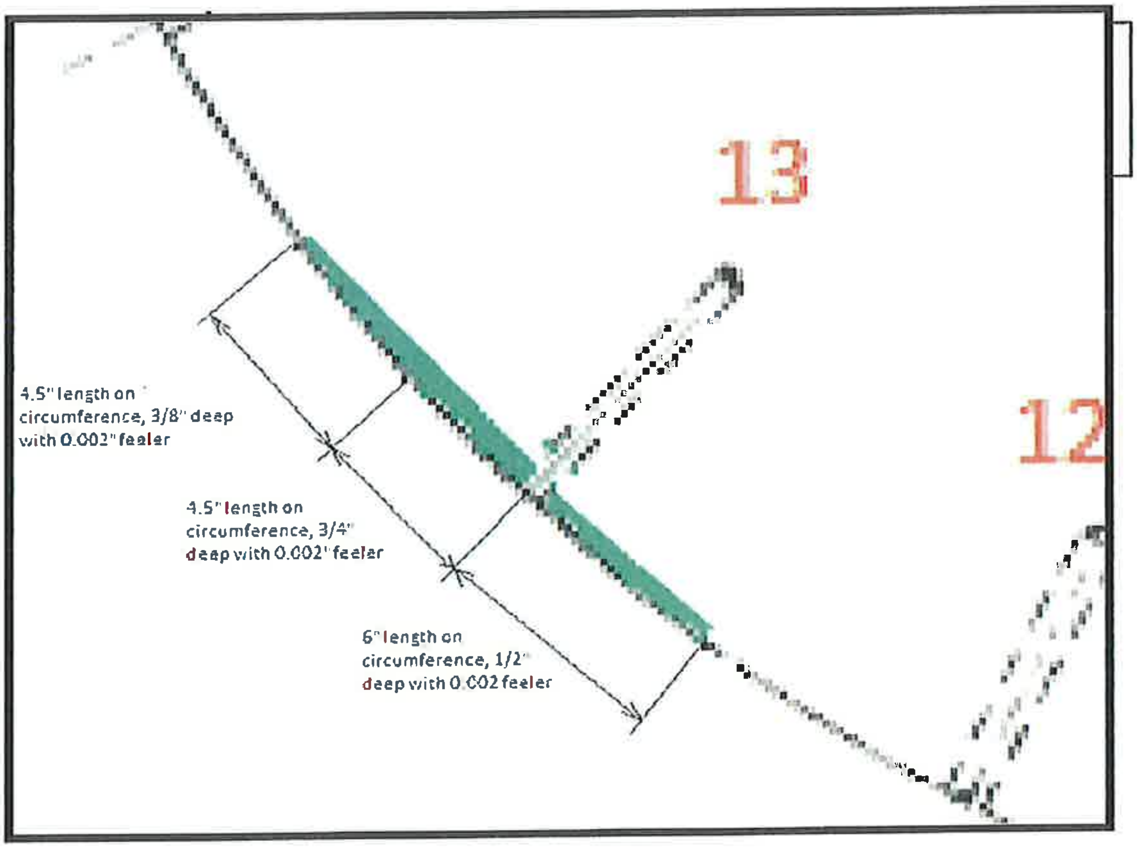
Revision # 00

ITP step #: 6301-7

Project: GM Shrum

Unit : 4

instruction : Measure zones that dont make contact between thrust bloc and coupling flange



NCR #: __________

Measurement Instruments	T°	Measured by: Mike Kelly	dd-mmm-yyyy
feeler gage	Before:		24-Jul-2012
	After:	ANDRITZ Rep.: Mike Kelly	
		Cust. Repres.: Katie Taylor	

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Q-0000-01

ITP step #: 6301-7

Project: GM Shrum

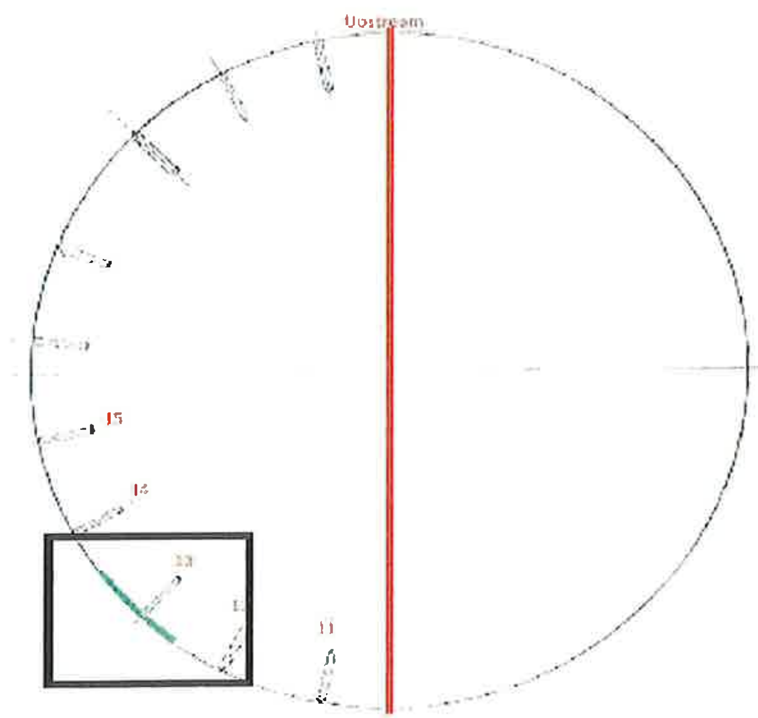
Unit : 4

Revision # 00

instruction : Measure zones that dont make contact between thrust bloc and coupling flange

NOTE:

While rotor was standing on stands and thrust block was brought as closely as possible without starting to torque, the biggest measurable gap was .008 in



NCR #: ______

Measurement Instruments	T°	Measured by: Mike Kelly	dd-mmm-yyyy
feeler gage	Before:		24-Jul-2012
	After:	ANDRITZ Rep.: Mike Kelly	
		Cust. Repres.: Katie Taylor	

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ITP step #: N/A
Project: GMS - rotor flange machining
Unit : 4
Date : july 22 2012

Equipment : Record rotor support level in the center of the support

Procedure P-GMS-6300-097 Step 1.3

point	laser tracker DRO elevation	added shim	added shim post rotor drop
1	44.093	0.078	
2	44.082	0.089	
3	44.171	0.000	
4	44.089	0.082	
5	44.101	0.070	
6	44.082	0.089	
7	44.121	0.050	0.048
8	44.058	0.113	0.098
9	44.118	0.053	
10	44.085	0.086	
11	44.156	0.015	
12	44.135	0.036	

On stand #8, additional shim was added on july 21 (.043in) using 2x 250T jack prior to rotor measurements

Machining step: Prior to rotor drop on stands
Time : Morning
Date : july 21 2012

NCR #: -

Measurement Instruments	T°	Measured by:	D. McCromac, D. Zaharoff	dd-mmm-yyyy
Laser tracker X0100802531	Before: 24.2 °C	ANDRITZ Rep.:	Mike Kelly	july 21 2012
	After: 25 °C	Cust. Repres.:	katie Taylor	

Installation and Commissioning

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Q-0000-001

ITP step #: N/A

Revision # 00

Project: GMS - rotor flange machining

Unit : 4

Page 1 of 1

Equipment : Record rotor support level in the center of the support

Procedure P-GMS-6300-097 Step 1.3

point	laser tracker DRO elevation	added shim	added shim post rotor drop
1	44.093	0.078	
2	44.082	0.089	
3	44.171	0.000	
4	44.089	0.082	
5	44.101	0.070	
6	44.082	0.089	
7	44.121	0.050	0.048
8	44.058	0.113	0.055
9	44.118	0.053	
10	44.085	0.086	
11	44.156	0.015	
12	44.135	0.036	

Machining step: Prior to rotor drop on stands

Time : Morning

Date : july 20 2012

NCR #:

Measurement Instruments		T°	Measured by:	P. Mitre D. MAccomac, D. Zaharoff	dd-mmm-yyyy
Laser tracker X0100802531	Before:	24.2			
	After:	24.5	ANDRITZ Rep.:	Mike Kelly	july 20 2012
			Cust. Repres.:	Katie Taylor	

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Q-0000-01

ITP step #: 6301-3

Revision # 00

Project: GM Shrum

Unit : 4

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Equipment : Thickness of rotor coupling flange after machining

#	ID	OD
1	3.419	3.384
2	3.406	3.376
3	3.412	3.377
4	3.437	3.396
5	3.460	3.417
6	3.462	3.423
7	3.447	3.407
8	3.443	3.398
9	3.441	3.396
10	3.447	3.406
11	3.443	3.396
12	3.452	3.408
13	3.460	3.419
14	3.466	3.402
15	3.447	3.403
16	3.433	3.390

Avg.	3.442	3.400
Max	3.466	3.423
Min	3.406	3.376

Note: * Measurement are taken in coupling holes using a straight edge and dept mic.

NCR #: ______

Measurement Instruments	T°		Measured by:	D. Zaharoff	dd-mmm-yyyy
dept micrometer	Before:	24.7		D.MacCormack	24-Jul-2012
#417050	After:	24.7	ANDRITZ Rep.:	Mike Kelly	
			Cust. Repres.:		